

Absolute Value

The absolute value of a number is its distance from zero, without considering its sign. This distance is always non negative, signifying the magnitude of a number.

Mathematically, the absolute value of a real number x is written as $|x|$. If $x \geq 0$, then $|x| = x$; if $x < 0$, then $|x| = -x$. For example, the absolute value of -5 is 5, while the absolute value of 7 is also 7. The absolute value concept is key in mathematics for comparing magnitudes, solving equations, measuring distances, and working with inequalities.

Example 5: Solve $|x + 1| = |2x - 3|$

To solve the equation $|x + 1| = |2x - 3|$, consider the following cases:

- Case 1: $x + 1 = 2x - 3$

$$x + 1 = 2x - 3 \Rightarrow 1 + 3 = 2x - x$$

$$4 = x \Rightarrow x = 4$$

- Case 2: $x + 1 = -(2x - 3)$

$$x + 1 = -2x + 3 \Rightarrow x + 2x = 3 - 1$$

$$3x = 2 \Rightarrow x = \frac{2}{3}$$

- Case 3: $-(x + 1) = 2x - 3$

$$-x - 1 = 2x - 3 \Rightarrow -x - 2x = -3 + 1$$

$$-3x = -2 \Rightarrow x = \frac{2}{3}$$

- Case 4: $-(x + 1) = -(2x - 3)$

$$-x - 1 = -2x + 3 \Rightarrow -x + 2x = 3 + 1$$

$$x = 4$$

Thus, the solutions to the equation $|x + 1| = |2x - 3|$ are:

$$x = \frac{2}{3} \text{ or } x = 4$$

Example 6: Solve $\frac{|x+2|}{|x+1|} > 0$

To solve it, it is necessary to understand when the fraction is positive.

For the fraction

$$\frac{|x+2|}{|x+1|}$$

to be greater than 0, both the numerator and the denominator must be non-zero and have the same sign. Since absolute values are always non-negative, the key is ensuring that neither $|x+2|$ nor $|x+1|$ is zero.

Cases to Consider:

- Numerator and denominator analysis:
 $|x+2| > 0$ and $|x+1| > 0$ when neither $x+2$ nor $x+1$ is zero.
- Identify Points Where the Absolute Values are Zero
 - $|x+2| = 0$ when $x = -2$
 - $|x+1| = 0$ when $x = -1$
- The fraction $\frac{|x+2|}{|x+1|}$ is undefined at $x = -1$ because division by zero is not allowed.
- At $x = -2$, $|x+2| = 0$, resulting in the entire fraction being zero, which does not satisfy the inequality

Therefore, the solution is $x \neq -2$ and $x \neq -1$

In the interval notation is:

$$x \in (-\infty, -2) \cup (-2, -1) \cup (-1, \infty)$$

Example 7: Solve $\frac{x^3+9x^2-36x}{|x^3-1|} \leq 0$,

Is necessary to analyze when the rational expression is less than or equal to zero. This involves several steps:

- Step 1: Factor the Numerator

The numerator is $x^3 + 9x^2 - 36x$. We can factor this polynomial as follows:

$$x^3 + 9x^2 - 36x = x(x^2 + 9x - 36)$$

Now, factor the quadratic expression $x^2 + 9x - 36$:

$$x^2 + 9x - 36 = (x + 12)(x - 3)$$

Thus,

$$x^3 + 9x^2 - 36x = x(x + 12)(x - 3)$$

So $\frac{x^3+9x^2-36x}{|x^3-1|} \leq 0$, turns into $\frac{x(x+12)(x-3)}{|x^3-1|} \leq 0$,

- Step 2: Analysis

- Undefined Points: The fraction is undefined at $x = 1$ because the denominator $|x^3 - 1|$ is equal to 0 there and division by zero is not allowed.

- Zero Points: The numerator is zero at $x = -12$, $x = 0$, and $x = 3$.

- Step 3: Find intervals where $\frac{x(x+12)(x-3)}{|x^3-1|} \leq 0$

1. Intervals to Consider:

We examine the signs of $x(x + 12)(x - 3)$ and $|x^3 - 1|$ around the critical points $x = -12$, $x = 0$, $x = 1$, and $x = 3$.

2. Sign Analysis:

For $x < -12$: The numerator is negative (since all factors x , $x + 12$, and $x - 3$ are negative), and the denominator is positive (since $x^3 - 1$ is negative and absolute value is positive). Thus, the fraction is negative.

For $-12 < x < 0$: The numerator is positive (since x is negative, but $(x + 12)$ and $(x - 3)$ are positive) and the denominator is positive. Thus, the fraction is positive.

For $0 < x < 1$: The numerator is negative (since x and $(x - 3)$ are negative) and the denominator is positive. Thus, the fraction is negative.

For $1 < x < 3$: The numerator is negative and the denominator is positive. Thus, the fraction is negative.

For $x > 3$: The numerator is positive (since all factors are positive) and the denominator is positive. Thus, the fraction is positive.

So in the interval notation the solution is:

$$x \in (-\infty, -12] \cup [0, 1) \cup (1, 3]$$